LAB GUIDELINES

Lab 19 — Physical and Environmental Security Walk-Through

Maps to CompTIA Server+ objective **3.2 — Summarize physical security concepts** (physical access controls: bollards, architectural reinforcements — signal blocking, reflective glass, datacenter camouflage; fencing; security guards; security cameras; locks — biometric, RFID, card readers; mantraps; safes; environmental controls — fire suppression, HVAC, sensors).

This is a hardware concept lab. Use the Killercoda VM only to **inspect environmental sensors** the OS can see; do the rest as a written walk-through.

Run all commands on https://killercoda.com/playgrounds/scenario/ubuntu

Required software (free, apt): `lm-sensors`, `smartmontools`

```
apt update && apt install -y lm-sensors smartmontools
```

Step 1 — Physical access control layers (defence in depth)

A real data centre stacks controls from outside in. Map each layer to its purpose:

Layer (outside → in)	Control	Purpose
Property perimeter	Fencing, bollards	Stop vehicles, channel pedestrians
Building	Datacenter camouflage	Avoid drawing attention
Lobby	Security guards, cameras	Detect + deter
Inner doors	Card reader + biometric (2-factor)	Authentication + audit
Trap between zones	Mantrap / airlock	Prevent tailgating
Server hall	Smart-card cabinets, safes for KMS / HSM	Last line of defence

Bollards = anti-ram posts. Architectural reinforcements = signal blocking (TEMPEST / Faraday), reflective glass, blast walls.

Step 2 — Locks the exam asks about

Lock type	Authenticator	Replay-resistant?
Mechanical key	Physical key	No — easy duplication
Card reader	Magstripe / proximity card	No — cards clonable
RFID	RFID badge (often + PIN)	Partly — challenge protocols
Biometric	Fingerprint, iris, face	Yes — but template needs protection

Best practice: **two-factor** — card + biometric, or card + PIN.

Step 3 — Environmental controls

```
sensors-detect --auto >/dev/null 2>&1 || true
sensors 2>/dev/null | head -30 || echo "no exposed sensors in this VM — on real hardware lm-sensors
smartctl -a /dev/sda 2>/dev/null | grep -i temperature | head -3 || true
```

Server+ 3.2 lists these environmental controls:

Control	Target
Fire suppression	Inert gas (FM-200, Novec 1230) — no water on electronics
HVAC	18–27 °C, 40–60 % RH per ASHRAE TC 9.9
Sensors	Temp, humidity, smoke, water-leak, door-open

Step 4 — Cameras and guards as detective controls

Control	Category
CCTV	Detective (and deterrent if visible)
Guard patrol	Detective + responder
Motion sensors	Detective
Door alarm	Detective
Visitor log	Detective + audit trail

Cameras retain footage for **30–90 days** typically (link to retention policy from Lab 17).

Step 5 — Datacenter physical zones

Draw four concentric rings:

1. **Perimeter** — fence, bollards, gatehouse, vehicle screening.
2. **Building shell** — windowless / reflective glass, signal-blocking walls.
3. **Operations area** — mantrap, NOC, break rooms, no servers.
4. **Server hall** — biometric + card, locked cabinets, hot/cold aisle (Lab 1).

Step 6 — Tailgating, piggybacking, and the mantrap

A mantrap admits **one person at a time**. The inner door cannot open until the outer door closes and the person inside passes a second factor. This defeats tailgating (unauthorised follow-through).

Step 7 — Sample physical-security checklist

```
[ ] Perimeter fence intact, gate alarms tested in last 90 days
[ ] CCTV coverage of every entry, NVR retention ≥ 30 days
[ ] Visitor badges colour-coded, escorted at all times
[ ] Card-reader access list reviewed quarterly
[ ] Biometric template database encrypted at rest
[ ] HVAC redundancy N+1, set 22 °C ±2, humidity 45 % ±10
[ ] Fire suppression inspected annually
[ ] Water-leak sensors under raised floor, alerts to NOC
[ ] Mantrap interlock tested monthly
[ ] Cabinet locks with audit log
```

Free online tools

- **Uptime Institute Tier standards** — https://uptimeinstitute.com/tiers
- **ASHRAE thermal guidelines** — https://www.ashrae.org/
- **NIST SP 800-116 (PIV / physical access)** — https://csrc.nist.gov/publications/sp

What you learned

- The full Server+ 3.2 vocabulary: bollards, architectural reinforcements, mantraps, RFID, biometric, signal blocking, reflective glass, datacenter camouflage.

- Lock and authenticator pros / cons, and why two-factor physical access matters.
- ASHRAE environmental targets and the sensor set a server room needs.
- Camera retention vs. policy.
- Concentric-ring data-centre zoning.
- A reusable physical-security audit checklist.